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ControlDesk - Basics

- HIL testing means the test of real ECUs in a realistic simulated environment. Usually a HIL test system comprises the units under test, which can be one or more ECUs containing new SW functions to be tested; real hardware components can be added to the test system.
- While the ECUs are accessible by the use of tools such as INCA, the simulated environment for HIL testing is created by ControlDesk platform.
- ControlDesk allows many features for the SW HIL testing:
 - ✓ simulate driving cycles and maneuvers in specific driving conditions to check the ECUs behaviors as for expectations;
 - ✓ monitoring the CAN frames for the real nodes in the loop;
 - ✓ IO check in open loop test and IO fault injection.



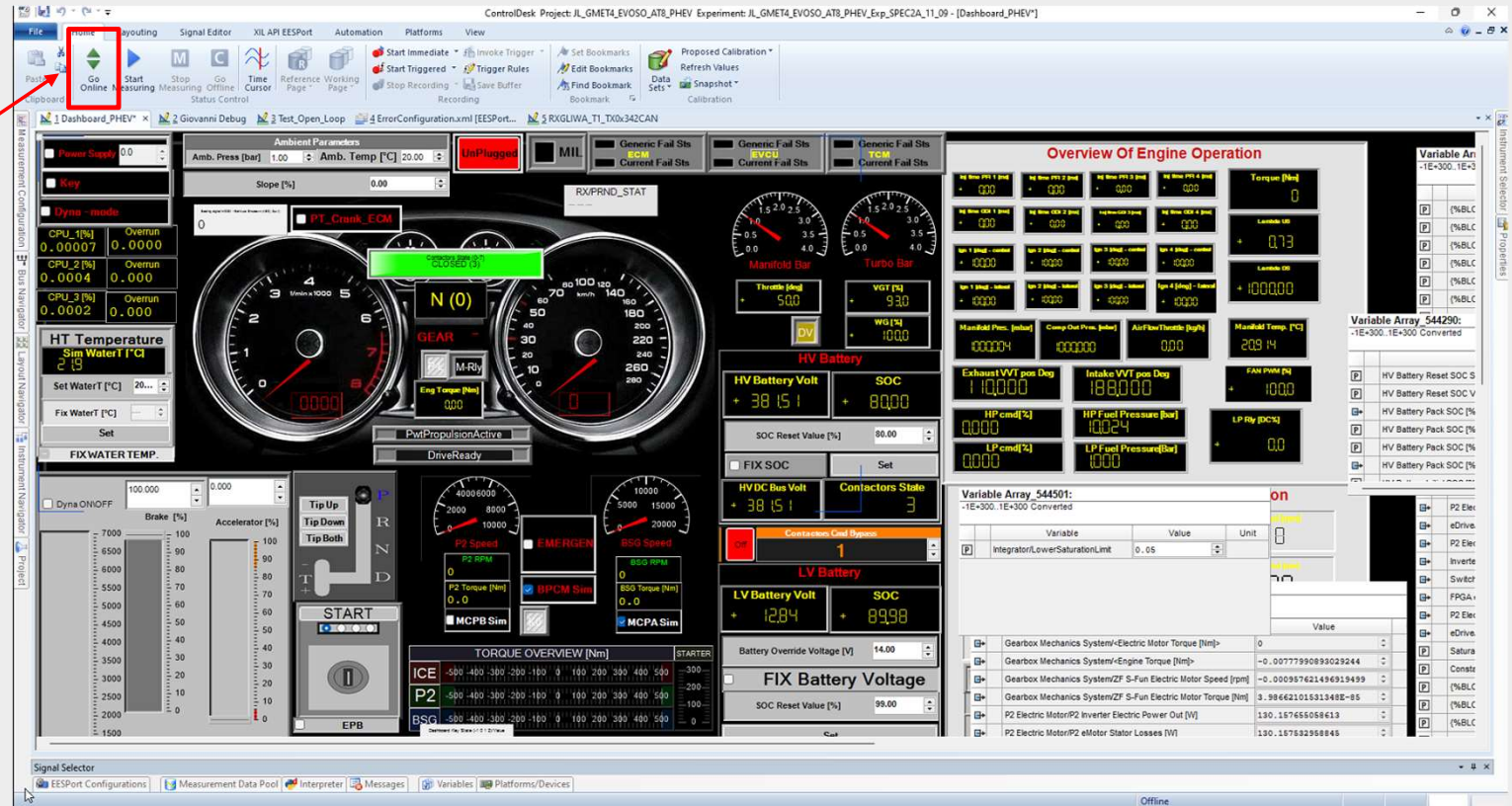
Model load and layout



- After opening the ControlDesk experiment, more than one .sdf file can be added in the *Project* section. You can choose the current .sdf file by activating it. The .sdf file contains all the variables description created in the engine model.
- If you don't know which .sdf to use, ask to the PL or the HIL team.

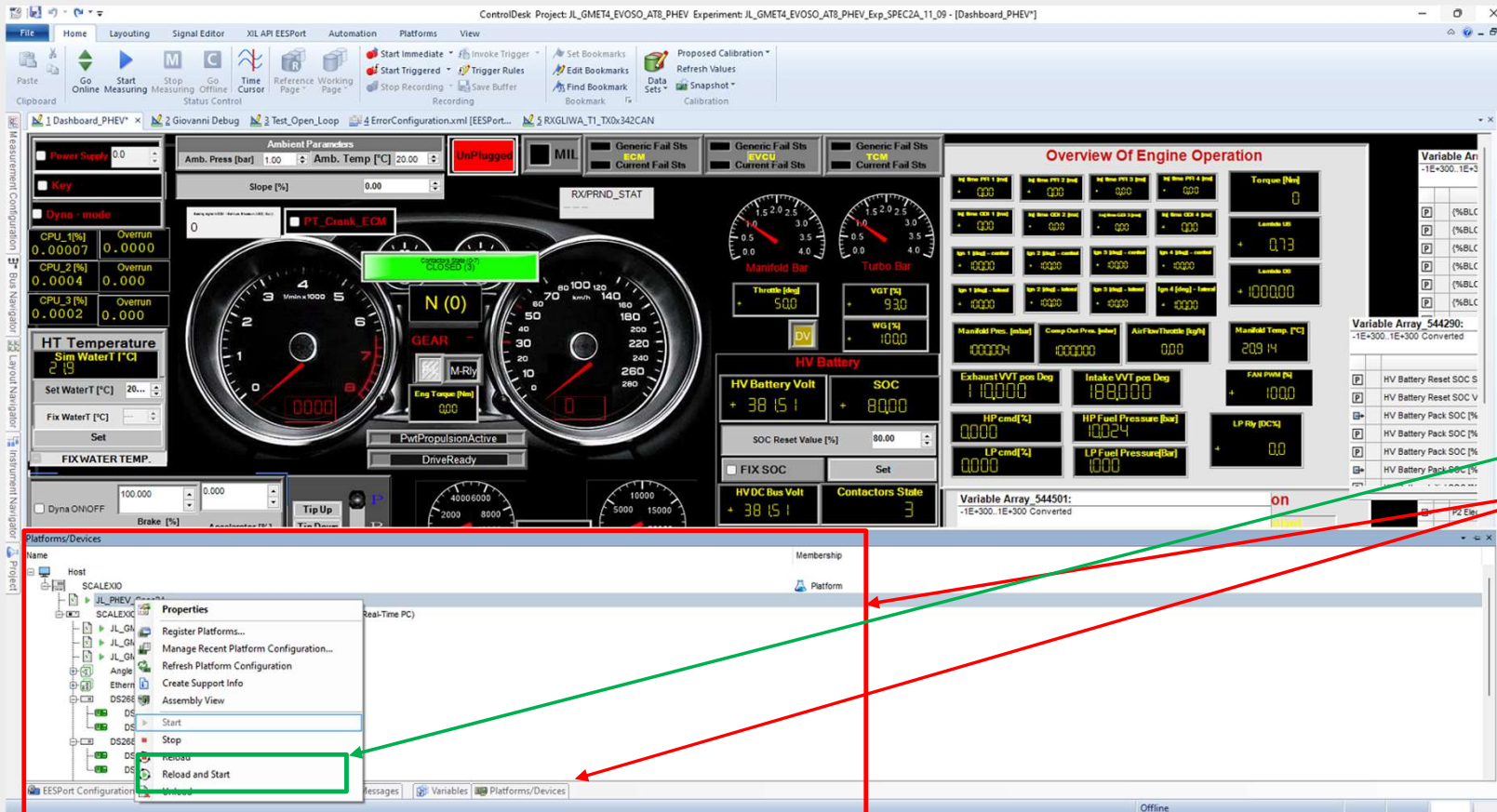
k Model load and layout

- Start the real time application by clicking **GO ONLINE** on the main view in ControlDesk. In this way the model and all its parameters are uploaded and ready to start.





Model load and layout



- Sometimes, it could be helpful to reload the model. In that case, starting from OFFLINE mode, choose the selection **Reload and Start** in the *Platform/Device* section.

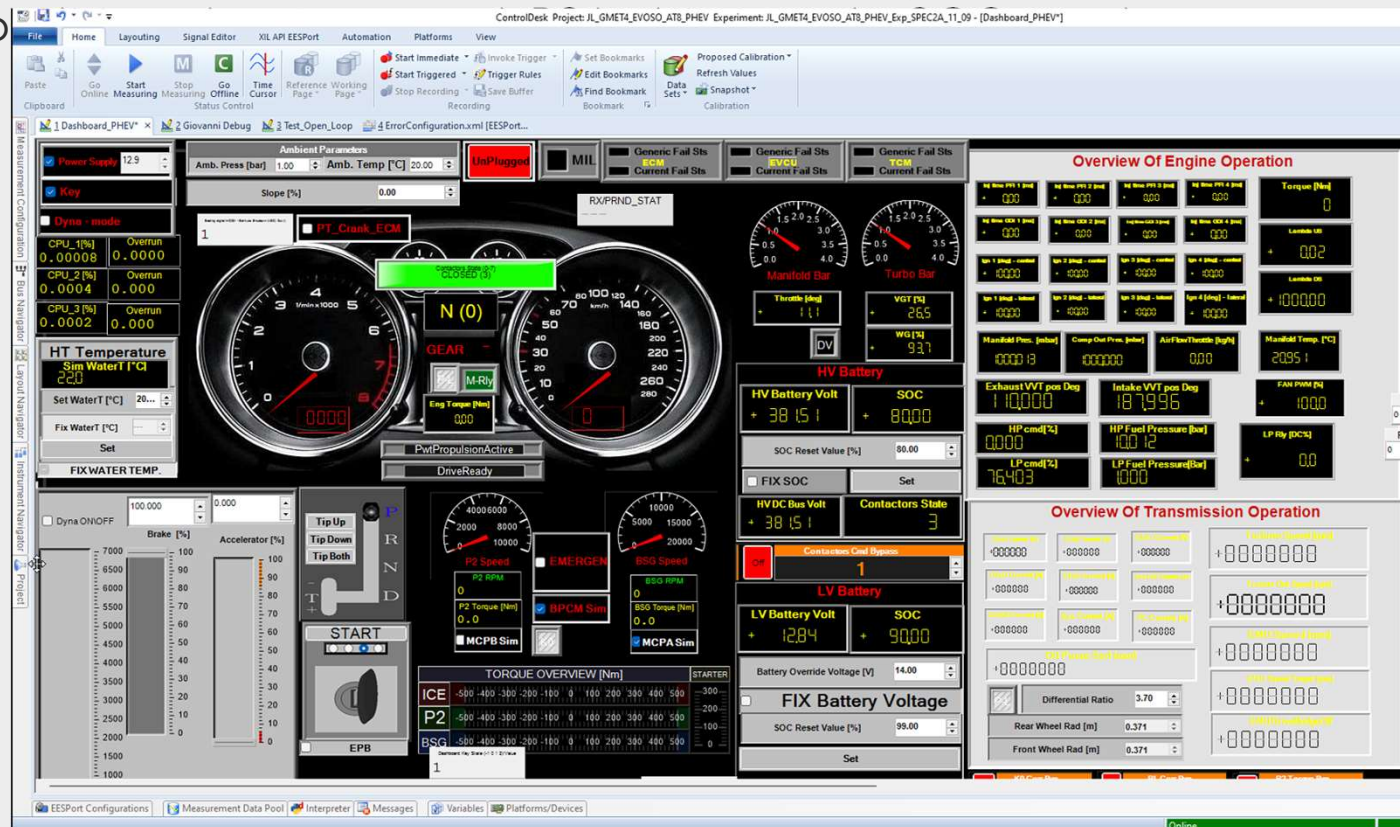


It's mandatory to reload the model after that the automation team use the HIL for its test.



Model load and layout

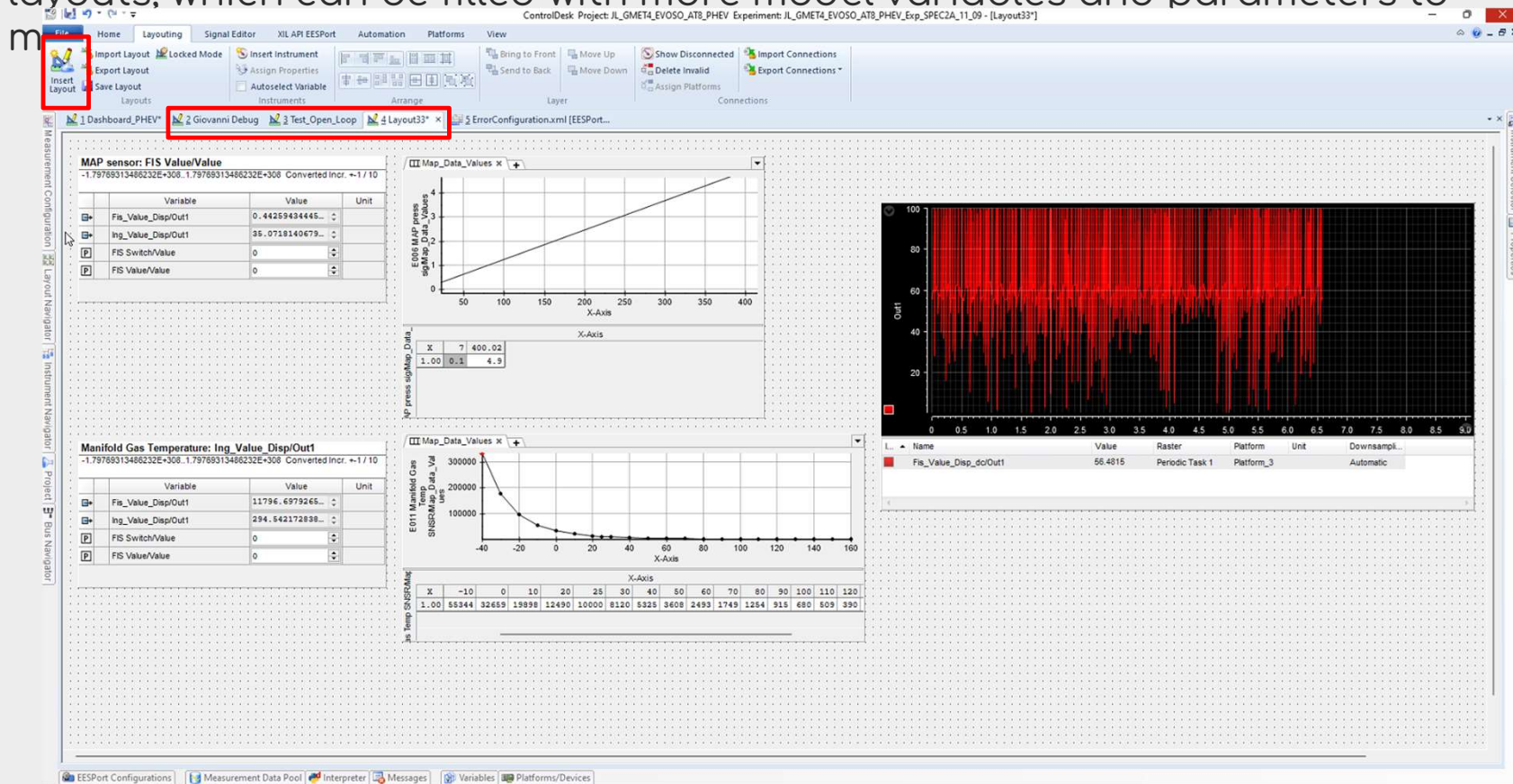
- Usually, the main layout (Dashboard) contains the most important parameters to set in the model and the commands to simulate the vehicle drive (key position, accelerator and brake pedal, gear shift, etc...). Many variables are added to show the driving co



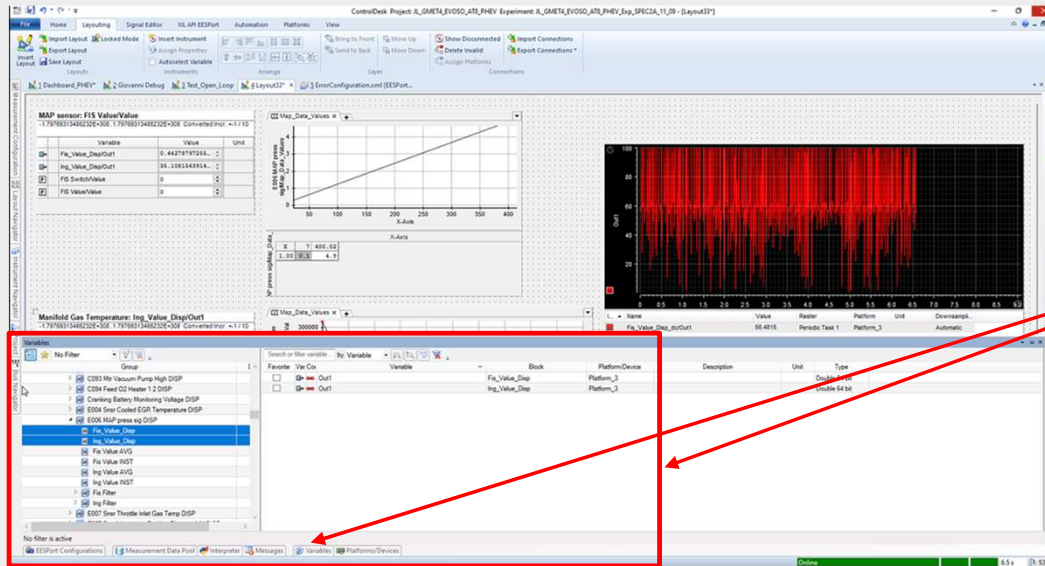


Model variables and IO signal display

- From the menu bar on the top in the section *Layouting*, you can create more layouts, which can be filled with more model variables and parameters to

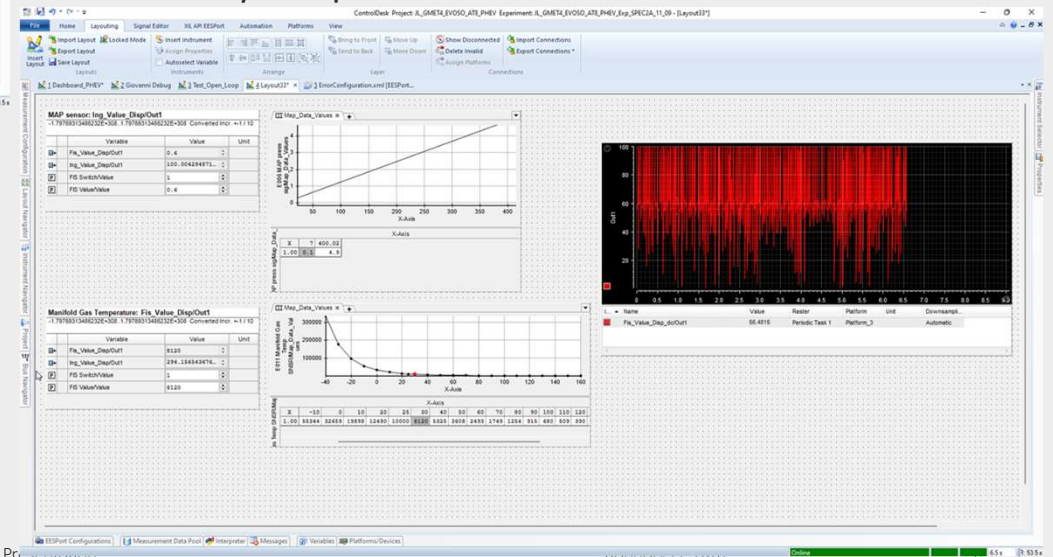


k Model variables and IO signal display

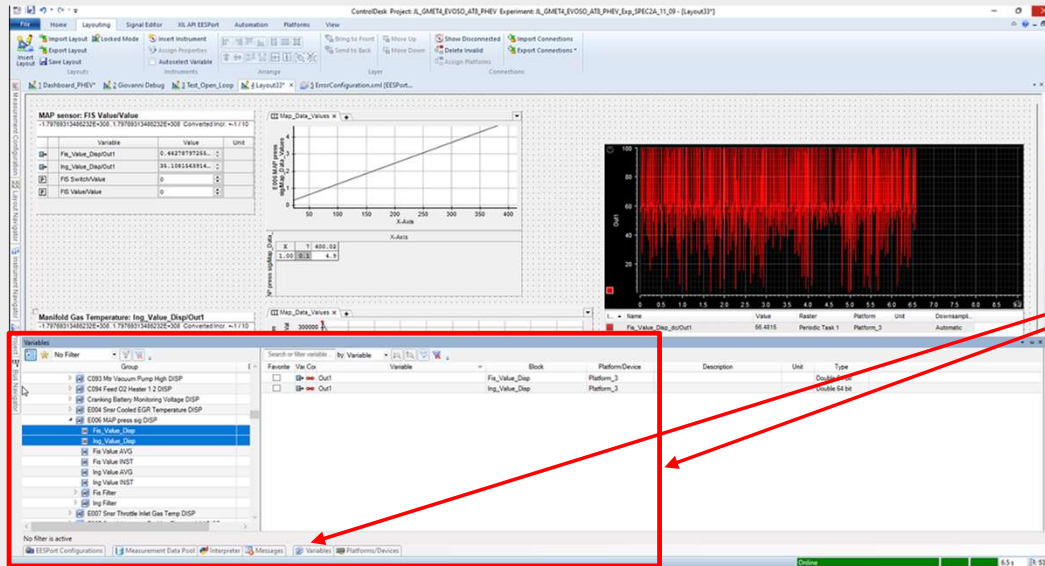


- To navigate into the model and select parameters and variables, you can go in the section *Variables*. You can add variable, dragging them into the layout, and display them as variable array or plot them.

- Adding the model variables can be useful to:
 - ✓ Check correspondence between the variables in the engine model and the ECU;
 - ✓ Open loop test on IO, most commonly with sensors and actuators;



k Model variables and IO signal display



- It is possible to override two variables with the same value at the same time. You drag the second variable in the window of the first one right clicking. Then choose the selection *Connect as Additional Write Variable*.

- To navigate into the model and select parameters and variables, you can go in the section *Variables*. You can add variable, dragging them into the layout, and display them as variable array or plot them.

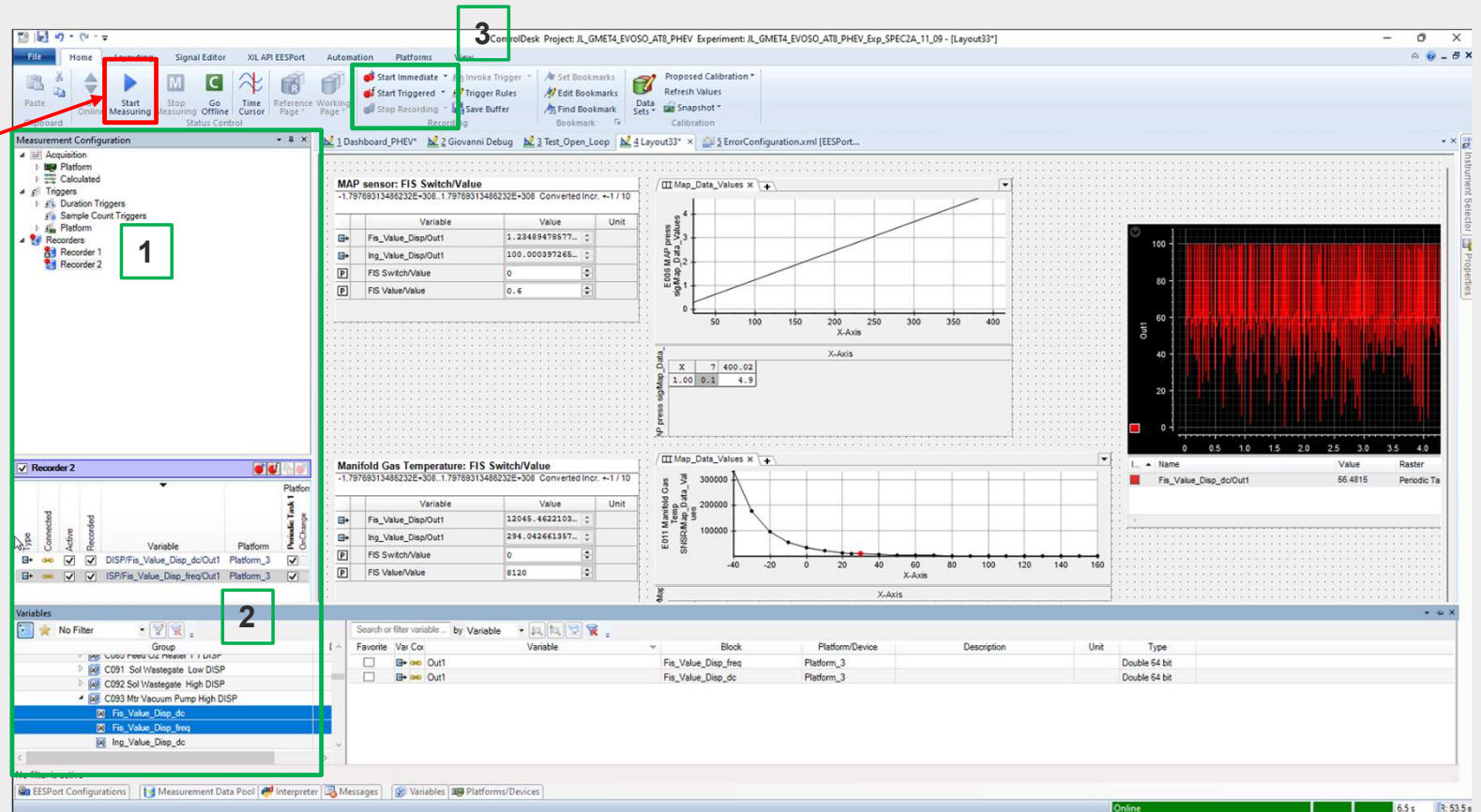
Variable Array_544513: ING Value/Value
-1.79769313486232E+308..1.79769313486232E+308 Converted Incr. +/- 10

	Variable	Value	Unit
[P]	FIS Switch/Value	0	
[P]	FIS Value/Value	0	
[P]	ING Switch/Value	0	
[P]	ING Value/Value	0	

Replace Variable
Insert Before
Append Variable
Connect as Additional Write Variable(s)
Cancel

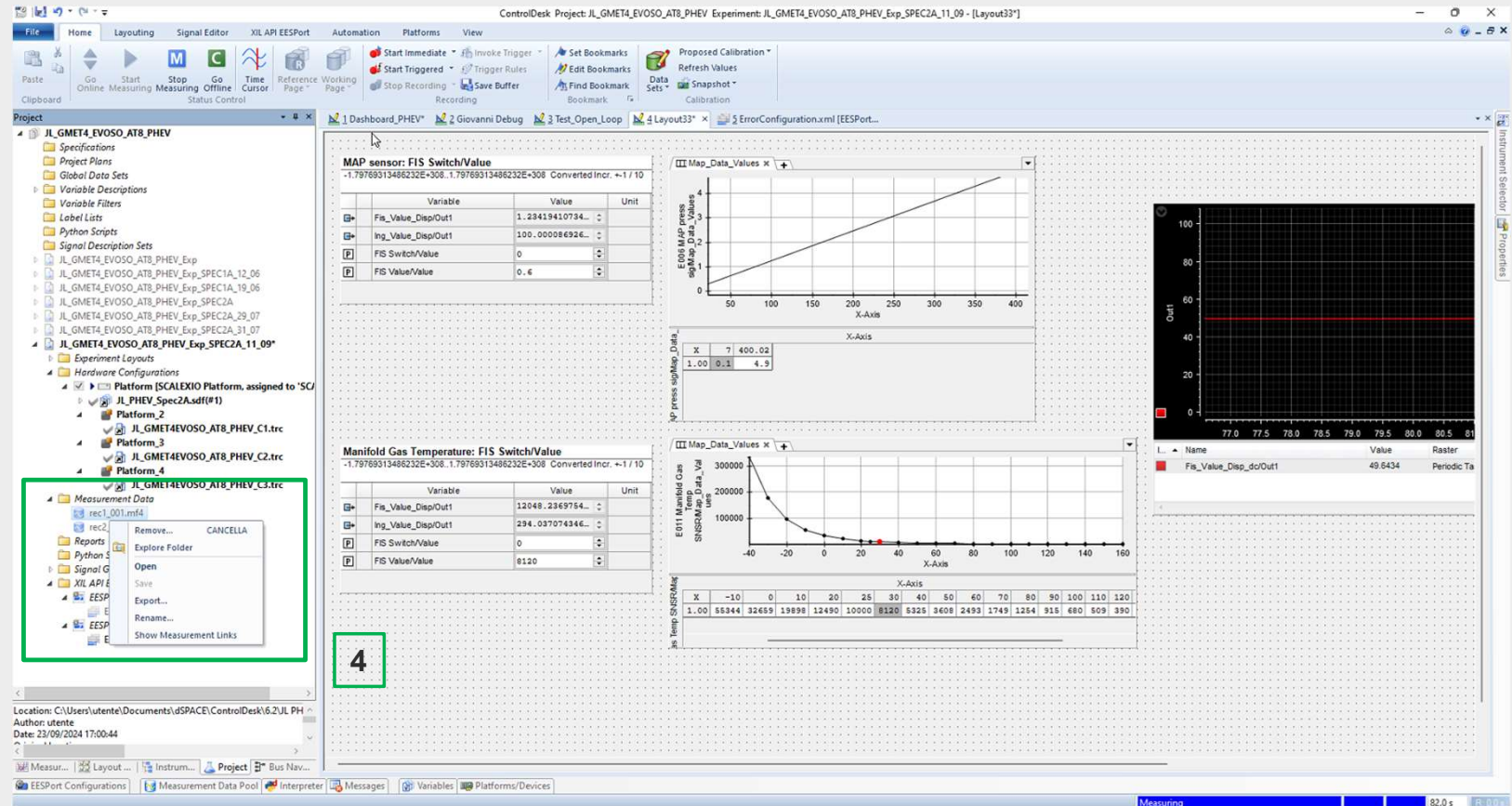
k Model variables and IO signal display

- When adding the model variables on the plot editor, you can measure the real time behavior. You can do that by clicking **Start Measuring** on the menu bar on the top.
- Sometimes, it could be helpful to record the model variables. You can do that following the procedure:
 - In *Measurement Configuration* section create a new recorder;
 - Select the variables to record and drag them in the recorder section;
 - Click on **Start immediate** to start the record and click on **Stop recording** to stop the log;
 - The record will be available in the section *Project -> Measurement data*. You can export it as .mf4 file.



k Model variables and IO signal display

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 - The record will be available in the section *Project* -> *Measurement data*. You can export it as .mf4 file.



k Bus Navigator and CAN monitoring

- In the section **Bus Navigator**, you can find the CAN messages implemented in the model. Some messages are from the simulated nodes, while other messages are from the real nodes in the loop.
- You can add the Bus Monitor and the CAN Loader.





Bus Navigator and CAN monitoring

- On the Bus Monitor, the whole flow of messages and signals can be monitored.

The screenshot displays the ControlDesk software interface. The top menu bar includes File, Home, Layouting, Signal Editor, XIL API EESPort, Automation, Platforms, and View. Below the menu is a toolbar with various icons for file operations, measurement, and automation. The main workspace is divided into several panels:

- Bus Navigator:** Located on the left, it shows a hierarchical tree of the system components. The tree is expanded to show the 'CAN' section, which includes 'CAN_11' and 'CAN_12'. 'CAN_12' is further expanded to show 'Monitor_1_CAN_12'.
- Bus Monitor:** The central panel displays a table of CAN messages. The table has columns for Identifier, Monitor Time, ID, Controller ID, Name, Dir, DLC, Data, and Delta Time. The data is filtered to show messages from the 'CAN_12' node.
- Bottom Panel:** Contains tabs for 'EESPort Configurations', 'Measurement Data Pool', 'Interpreter', 'Messages', 'Variables', and 'Platforms/Devices'. The 'Measurement Data Pool' tab is active, showing a progress bar and a status indicator.

The Bus Monitor table shows a list of messages with their identifiers, times, and names. The messages are filtered to show only those from the 'CAN_12' node. The bottom panel shows the 'Measurement Data Pool' tab with a progress bar and a status indicator.



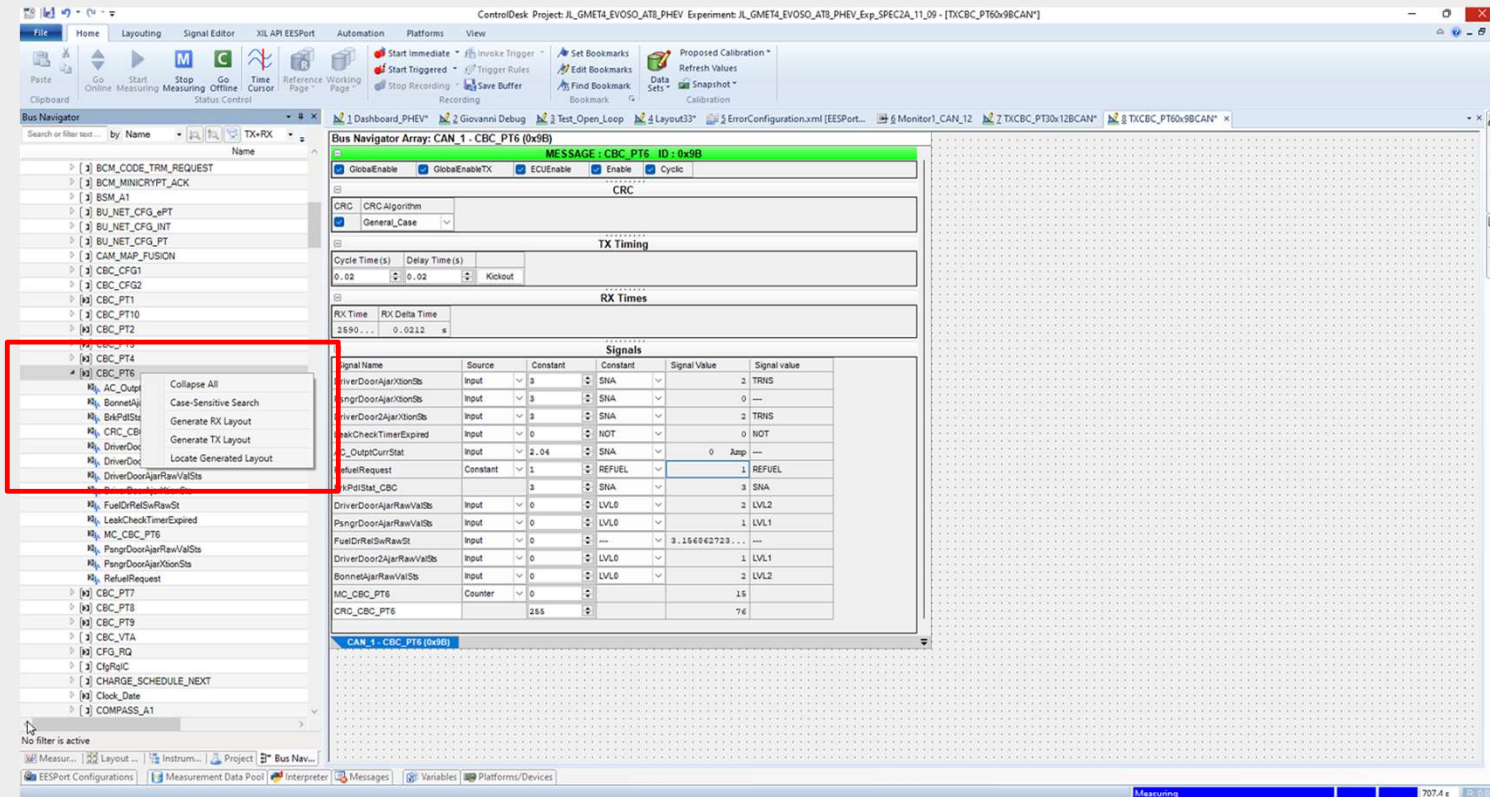
Bus Navigator and CAN monitoring

- It is possible to record the CAN frames by adding the CAN Logger.

The screenshot displays the Kineton ControlDesk software interface. The 'Bus Navigator' panel on the left shows a tree view of the system components, including 'Platform_2[1]', 'RT-CANHW', 'C1_Current', 'CAN_11', 'CAN_12', and 'Logger1_CAN_12'. The 'Monitor1_CAN_12' window is active, showing a list of CAN frames with columns for Identifier, Monitor Time, ID, Controller ID, Name, Dir, DLC, Data, and Delta Time. A context menu is open over the 'Monitor1_CAN_12' entry, with options: Case-Sensitive Search, Start, Delete, Activate (selected), and Edit. The bottom status bar indicates 'Measuring' and '532.9 s'.

Identifier	Monitor Time	ID	Controller ID	Name	Dir	DLC	Data	Delta Time
0x306 (0x3, Rx, 1)	5.6619999	0x306	3	BPCM_HV_SOC	Rx	8	00 00 00 00 00 00 0C C7	0.0999046
0x082 (0x3, Tx, 1)	5.6708866	0x082	3	Brake_Axle_Torque2	Tx	8	05 F4 05 F4 05 F4 05 05	0.0098072
0x094 (0x3, Rx, 1)	5.6754974	0x094	3	ESTorqueReq	Rx	8	04 80 00 03 FF 00 D0 B3	0.0200145
0x106 (0x3, Tx, 1)	5.6711226	0x106	3	ESC_EPT1	Tx	8	40 00 00 00 00 00 90 3F	0.0098072
0x486 (0x3, Rx, 1)	5.4322124	0x486	3	EV_RTC_FRAME	Rx	8	38 CD 9E 40 00 00 00 00	1.0001860
0x302 (0x3, Rx, 1)	5.4344048	0x302	3	Engine_Status5	Rx	8	00 00 00 00 00 00 00 00	0.0999462
0x08A (0x3, Rx, 1)	5.6745250	0x08A	3	Engine_Status7	Rx	8	03 AA 00 00 00 00 00 C5	0.0099789
0x18E (0x3, Rx, 1)	5.6747450	0x18E	3	Engine_Torque_Capacity2	Rx	8	00 00 00 00 00 00 80 2C	0.0099928
0x191 (0x3, Rx, 1)	5.6750090	0x191	3	Engine_Torque_Status	Rx	8	00 00 00 00 03 5A 80 1F	0.0099928
0x2A4 (0x3, Rx, 1)	5.6617497	0x2A4	3	HCP_DriverInfo	Rx	8	00 00 00 00 00 00 00 00	0.0999046
0x098 (0x3, Rx, 1)	5.6752531	0x098	3	HCP_TORQ_STATUS_FW	Rx	8	3E 83 83 3E 83 83 0B 0B	0.0099448
0x099 (0x3, Rx, 1)	5.6759858	0x099	3	HCP_TORQ_STATUS_FW2	Rx	8	3E 83 83 00 00 00 88 39	0.0104753
0x0EE (0x3, Rx, 1)	5.6794051	0x0EE	3	Hybrid_Axle_Sum_Torque1	Rx	8	00 00 00 00 00 00 00 00	0.0104632
0x0FE (0x3, Rx, 1)	5.6798534	0x0FE	3	Hybrid_Axle_Torque2	Rx	8	00 00 00 00 00 00 00 00	0.0104632
0x0A9 (0x3, Rx, 1)	5.6762301	0x0A9	3	Hybrid_Eng_Torq_Status	Rx	8	00 00 00 00 00 00 00 FF	0.0102310
0x0C4 (0x3, Rx, 1)	5.6781500	0x0C4	3	Hybrid_P2_Status1	Rx	8	07 D0 03 7D 80 00 00 F3	0.0104693
0x0C6 (0x3, Rx, 1)	5.6784284	0x0C6	3	Hybrid_P2_Status2	Rx	8	3F FF 03 E8 07 D0 B0 31	0.0104693
0x0C2 (0x3, Rx, 1)	5.6776756	0x0C2	3	Hybrid_PX_Status1	Rx	8	40 00 90 10 13 FF 80 85	0.0112020
0x0B4 (0x3, Rx, 1)	5.6764764	0x0B4	3	Hybrid_Status1	Rx	7	00 00 00 00 00 00 9A	0.0102310
0x1D7 (0x3, Rx, 1)	5.6698867	0x1D7	3	Hybrid_Status2	Rx	8	00 00 00 00 00 00 A0 AB	0.0092995
0x301 (0x3, Rx, 1)	5.6615765	0x301	3	Hybrid_Status3	Rx	8	00 00 FE 00 00 00 70 D6	0.1007463
0x08E (0x3, Rx, 1)	5.6793610	0x08E	3	MCPB_HYBRID_R1	Rx	8	00 00 AD 90 00 00 00 00	0.0104633
0x283 (0x3, Rx, 1)	5.6608294	0x283	3	MCPB_HYBRID_R2	Rx	8	00 00 79 00 00 00 00 00	0.0992445
0x097 (0x3, Rx, 1)	5.6757376	0x097	3	PT_SYSTEM_EPT_1	Rx	8	00 00 00 00 00 00 00 00	0.0099468
0x113 (0x3, Rx, 1)	5.6801057	0x113	3	ReGenBkStat2	Rx	8	00 00 00 00 00 00 80 88	0.0104632
0x210 (0x3, Tx, 1)	5.6615613	0x210	3	SBW_ROT1_DPT	Tx	5	01 00 00 20 43	0.0199097
0x093 (0x3, Rx, 1)	5.6767005	0x093	3	TRNS_STAT_DPT	Rx	8	00 00 4E 00 00 00 42 AA	0.0200125
0x09D (0x3, Rx, 1)	5.6774352	0x09D	3	Trans_Gen4_Status2	Rx	8	02 00 07 FE 7F 7F 80 E0	0.0102272
0x161 (0x3, Rx, 1)	5.6803519	0x161	3	Trans_Gen4_Status3	Rx	8	08 00 54 00 00 00 00 00	0.0200109
0x096 (0x3, Rx, 1)	5.6769447	0x096	3	Trans_Gen4_Status5	Rx	8	00 00 00 00 01 F4 80 4C	0.0102290
0x0C7 (0x3, Rx, 1)	5.6788685	0x0C7	3	Trans_P2_Status1	Rx	8	03 E8 00 00 00 00 80 19	0.0104593
0x0C9 (0x3, Rx, 1)	5.6791147	0x0C9	3	Trans_P2_Status2	Rx	8	00 00 EF 10 00 00 00 04	0.0104613
0x0C3 (0x3, Rx, 1)	5.6773098	0x0C3	3	Trans_PX_Status1	Rx	8	00 00 00 00 01 E2 81 70	0.0104633
0x0C5 (0x3, Rx, 1)	5.6783901	0x0C5	3	Trans_PX_Status2	Rx	8	00 00 00 64 20 00 80 8C	0.0104613
0x09C (0x3, Rx, 1)	5.6771870	0x09C	3	Trans_PX_Status4	Rx	8	00 00 1F 40 00 00 00 00	0.0102272

Bus Navigator and CAN monitoring



- In case of simulated nodes, you can override the signals. You can *Generate a TX layout* for a specific message and fill the values of the signals, changing it from input to constant.
- You can also:
 - ✓ disable the message to simulate a missing message, by deselecting the Global Enable on the top of the layout;
 - ✓ change the CRC by deselecting it, to simulate a fail on the CRC;
 - ✓ change the message counter, to simulate a fail on the MC.

k FIU (Fault Injection Unit)

- On the top bar, in the section *XIL API ESSPort*, you can configure and add a new layout for the FIU. Below the step procedure:
 - Configure the ESSPort;
 - Select the pin corresponding to the sensor/actuator/CAN on which you want to simulate the fault. The list is visible in the section ESSPort Configurations on the bottom and all the available fault on the specific pin are marked;
 - Drag the variable from the list in the spot;
 - Choose and modify the fault type (SCG, SCB, OL) in the section Properties;
 - Download, activate and trigger the event.
 - To heal the fault, click on unload and deactivate

1

2

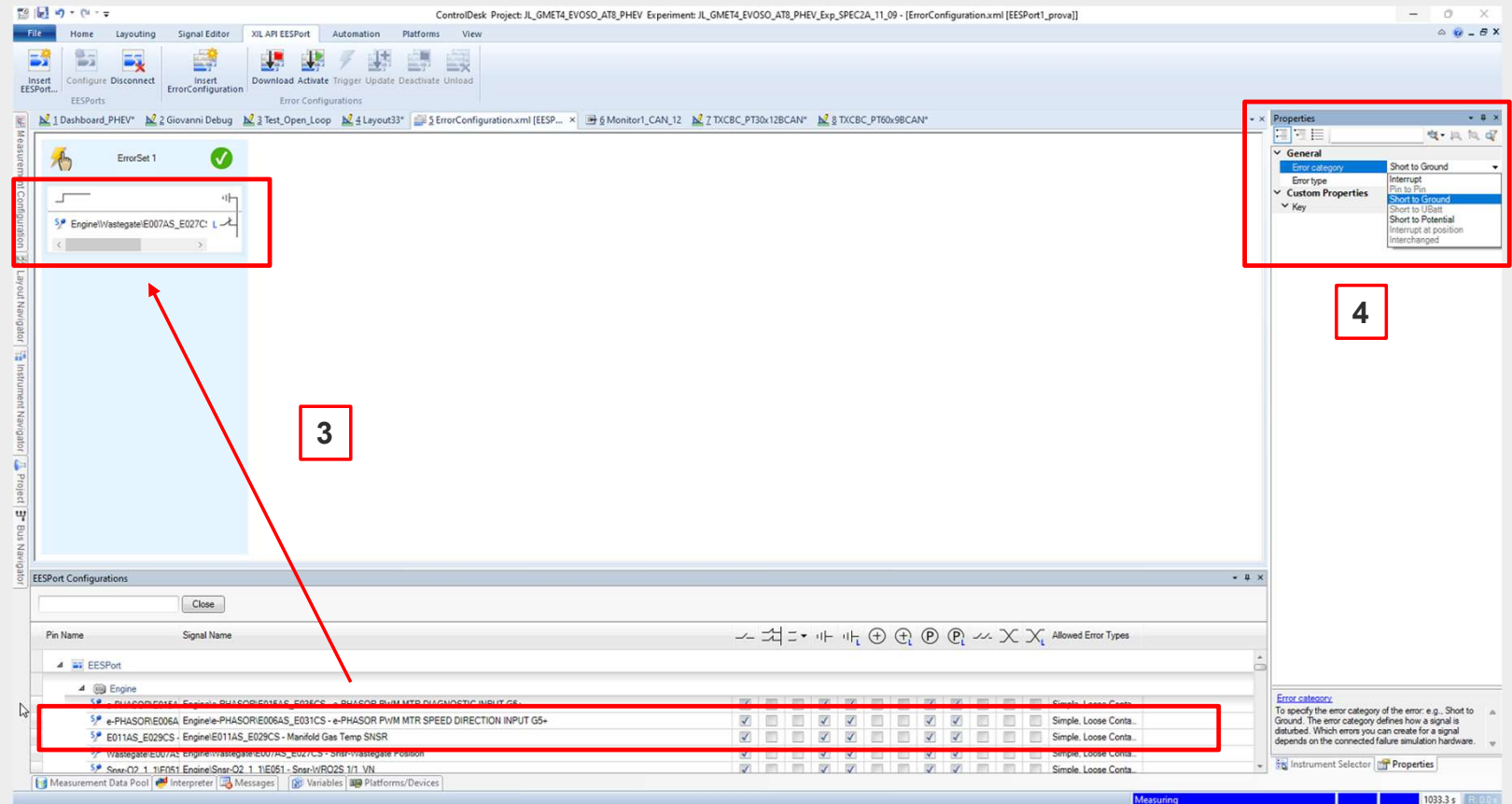
The 'ESSPort Configurations' dialog is shown, displaying a table of pins and their associated signals. The table has columns for 'Pin Name', 'Signal Name', and 'Allowed Error Types'. The 'Allowed Error Types' column includes checkboxes for 'Simple', 'Loose', and 'Cont.'. The table lists several pins and their corresponding signals, such as 'e-PHASORIE015A', 'e-PHASORIE006A', 'E011AS_E029CS', 'Wastegate/E007AS', 'Snar-Q2_1_1E051', and 'Snar-Q2_1_1E062'.

Pin Name	Signal Name	Simple	Loose	Cont.
e-PHASORIE015A	Engine-e-PHASORIE015AS_E035CS - e-PHASOR PWM MTR DIAGNOSTIC INPUT G5+	☒	☐	☐
e-PHASORIE006A	Engine-e-PHASORIE006AS_E031CS - e-PHASOR PWM MTR SPEED DIRECTION INPUT G5+	☒	☐	☐
E011AS_E029CS	Engine/E011AS_E029CS - Manifold Gas Temp SNSR	☒	☐	☐
Wastegate/E007AS	Engine/Wastegate/E007AS_E027CS - Snar-Wastegate Position	☒	☐	☐
Snar-Q2_1_1E051	Engine/Snar-Q2_1_1E051 - Snar-WRO2S 1/1_VN	☒	☐	☐
Snar-Q2_1_1E062	Engine/Snar-Q2_1_1E062 - Snar-WRO2S 1/1_VG	☒	☐	☐

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1. Configure the ESSPort;
2. Select the pin corresponding to the sensor/actuator/CAN on which you want to simulate the fault. The list is visible in the section ESSPort Configurations on the bottom and all the available fault on the specific pin are marked;
3. Drag the variable from the list in the spot;
4. Choose and modify the fault type (SCG, SCB, OL) in the section Properties;
5. Download, activate and trigger the event.
6. To heal the fault, click on unload and deactivate



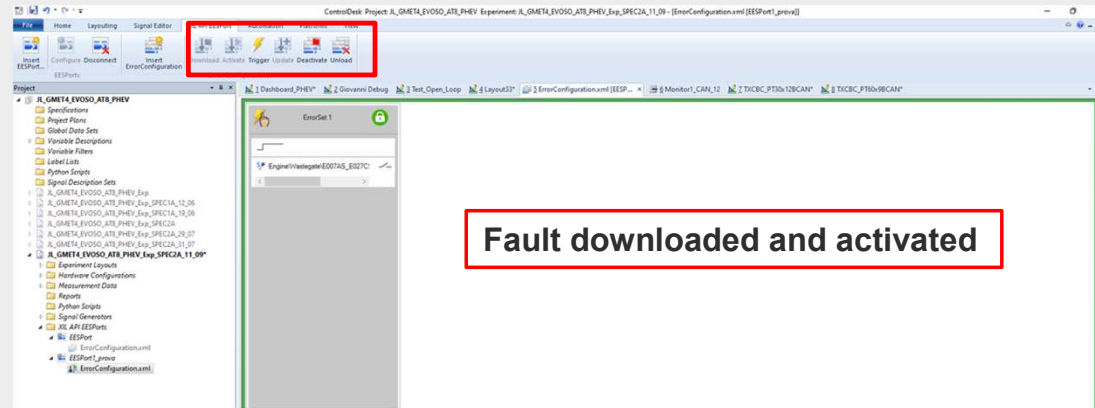
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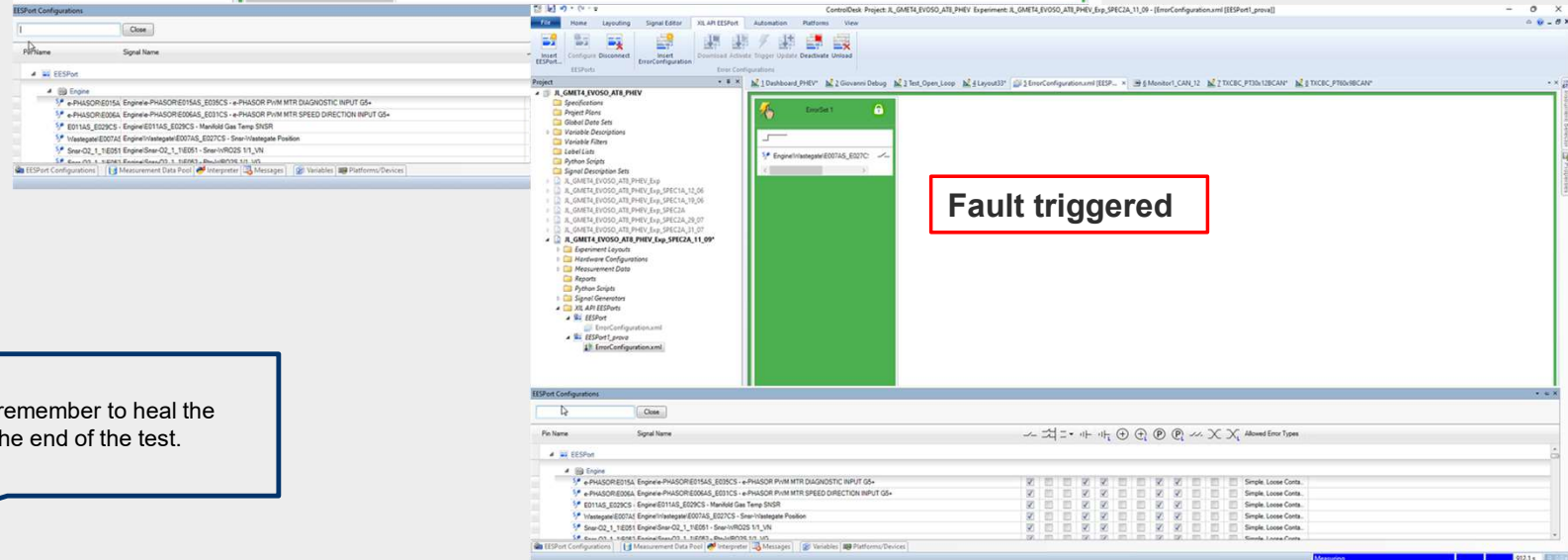
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3. Drag the variable from the list in the spot;
4. Choose and modify the fault type (SCG, SCB, OL) in the section Properties;
5. Download, activate and trigger the event.
6. To heal the fault, click on unload and deactivate



Always remember to heal the fault at the end of the test.

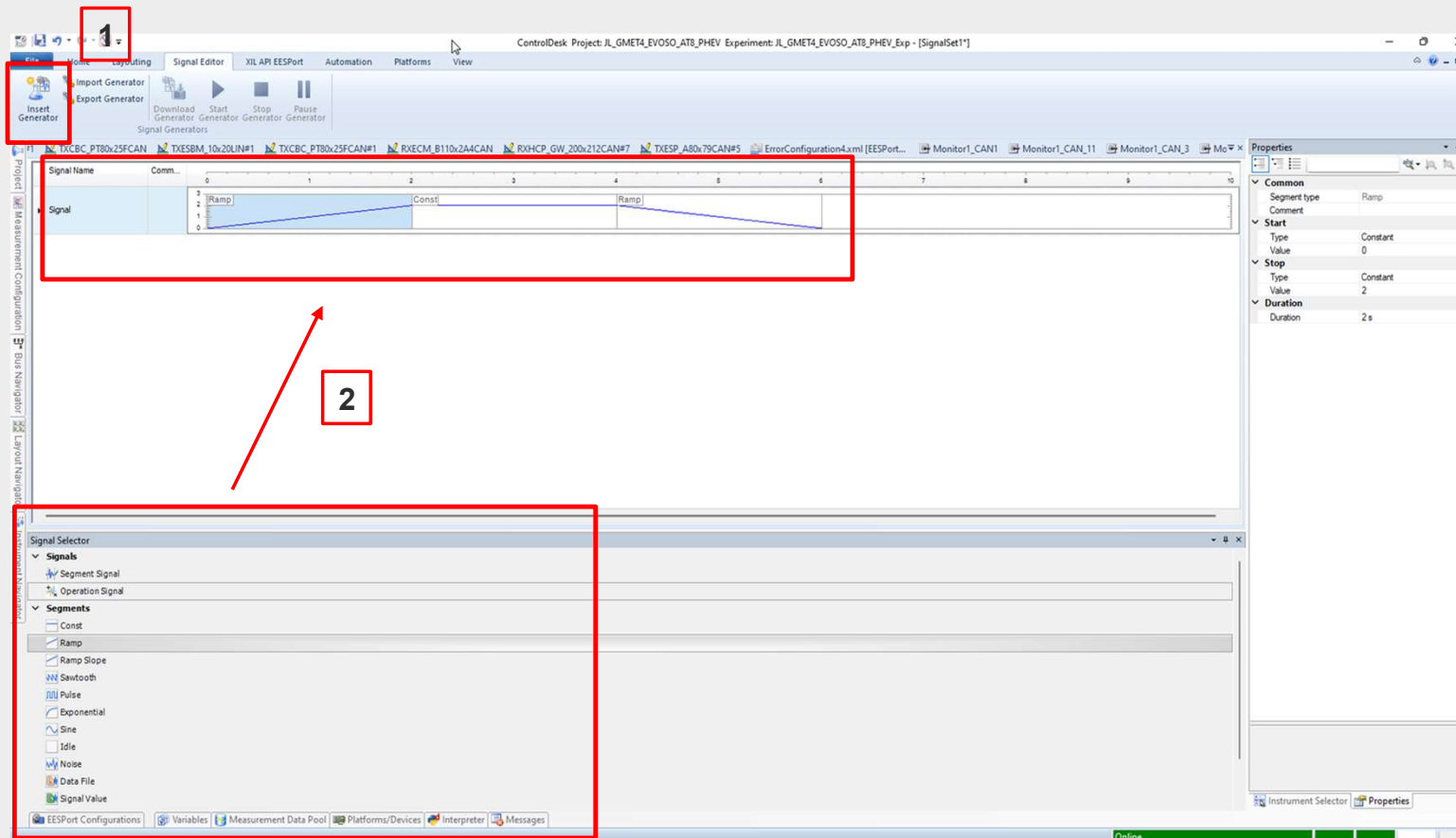


Fault downloaded and activated



Fault triggered

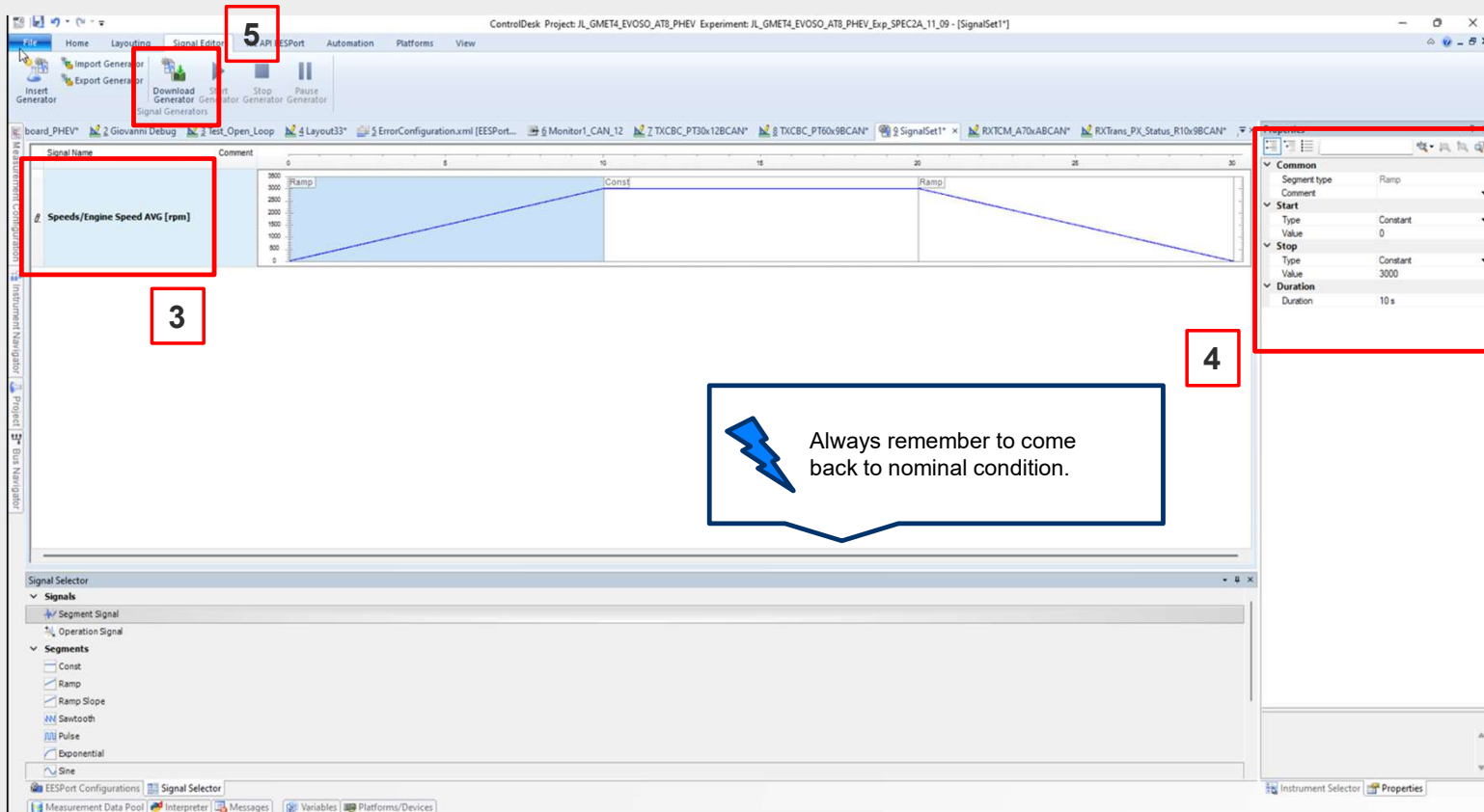
Signal generator



- On the top bar, in the section *Signal Generator*, you can generate a specific profile for a model variable (for example engine speed ramp, cycling key off-key on, etc.). Below the step procedures:

1. Click on Insert Signal Generator;
2. In Signal selector, choose the type of signal that you want to generate (segment signal) and drag into the bar the chosen segment;
3. Select the variable from the model and drag it into the spot;
4. Choose and modify the duration/period of the signal in the section Properties;
5. Download and start the Generator to trigger the signal.

Signal generator



The screenshot shows the Kineton Signal Generator interface. A red box labeled '5' highlights the top toolbar, specifically the 'Insert Signal Generator' button. Another red box labeled '3' highlights the 'Signal Name' field, which contains the text 'Speeds/Engine Speed AVG [rpm]'. A third red box labeled '4' highlights the 'Properties' panel on the right, which shows settings for a 'Ramp' segment, including 'Start' (Type: Constant, Value: 0), 'Stop' (Type: Constant, Value: 3000), and 'Duration' (10 s). A blue lightning bolt icon points to a text box that says 'Always remember to come back to nominal condition.' The main workspace shows a signal graph with a ramp segment and a constant segment.

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1. Click on Insert Signal Generator;
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3. Select the variable from the model and drag it into the spot;
4. Choose and modify the duration/period of the signal in the section Properties;
5. Download and start the Generator to trigger the signal.



Best practises

- Before starting using Control desk, check which project/experiment is activated in the Project section. If you don't know which is the correct .sdf to use, ask to your PL or HIL team.
- Before starting using Control desk, it is recommended to reload the experiment; it is mandatory to do it after an automation test.
- Before starting using Control desk, check that it is ONLINE and the Power Supply in ON.
- If you override some parameters or variables by Control Desk (for example override the voltage on a sensor), then remember to come back to the nominal state after your testing.
- If you change the value of some signals on Bus Navigator, then remember to come back to the nominal state after your testing.
- If you apply some faults with FIU, then remember to heal the fault and to come back to the nominal state after your testing.
- If you apply a stimulus with Signal generator, then remember to come back to the nominal state after your testing.
- Before leaving the HIL, always key OFF, switch the Power Supply OFF and go OFFLINE.